DisCo-FEND: Social Context Veracity Dissemination Consistency-Guided Case Reasoning for Few-Shot Fake News Detection

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Abstract. With the rapid development of the Internet, traditional news channels are being supplanted, leading to an increased prevalence of fake news. Mainstream pre-trained language models (PLMs)-based fake news detection methods follow the 'pre-training and fine-tuning' paradigm, relying on full supervision and heavily dependent on large, high-quality datasets. In contrast to these methods, "pre-trained and prompt-tuning" offers more efficient learning, especially in data-scarce scenarios. Meanwhile, extensive analysis of social patterns reveals a tendency driven by user psychology and behavior: users often disseminate information that aligns with their pre-existing beliefs, thereby reinforcing and solidifying their convictions. This phenomenon is termed "social context veracity dissemination consistency". Inspired by this phenomenon, we propose DisCo-FEND, A social context veracity Dissemination Consistency-guided case reasoning augmentation for the Fake News Detection (FEND) task. During model inference, we adopt a novel strategy that enhances reasoning by using multiple FEND cases. It leverages multiple news cases with higher dissemination consistency to refine predictions. Additionally, a high-quality label words acquisition approach and an adaptive weight allocation-based multi-label words mapping strategy improves the convergence and generalization of DisCo-FEND.

Keywords: Few-shot fake news detection · Social Context Veracity Dissemination Consistency Network · User spreading news engagement bias · Prompt-tuning · Case-based reasoning augmentation.

1 Introduction

With the pervasive use of the Internet, accessing external information has become significantly more convenient, greatly enriching our daily lives [3]. However,

the prevalence of fake news on the internet, which distorts facts, often leads social media users to adopt distorted thoughts and incorrect stances [14]. Fake news detection (FEND), serves to determine whether a news article is intentionally or unintentionally misleading, regardless of its topic or domain, through automated technology [14, 7]. Research into FEND is gaining increasingly attention from both the scientific community and government security agencies for a decade now.

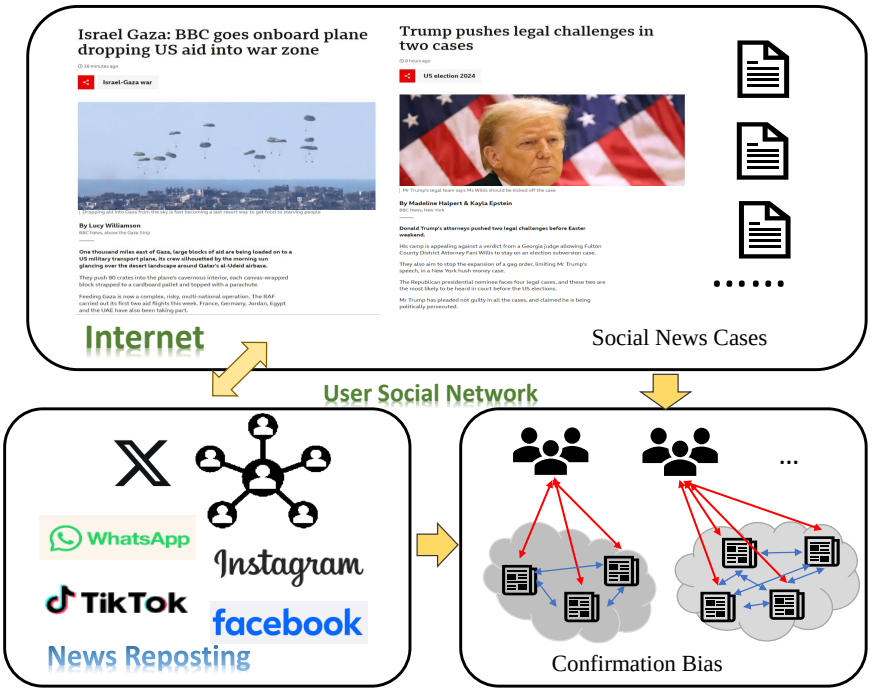


Fig. 1. A concept illustration of the social phenomenon: Confirmation Bias. This figure demonstrates how confirmation bias leads to the separation of cohorts based on their pre-existing beliefs. It shows how individuals selectively engage with news that reinforces their views, resulting in distinct partitions of information consumption.

Confirmation bias is a common social phenomenon propelled by cognitive tendencies and ideological biases [9]. Recently, researchers [11, 14] examined this phenomenon from psychological and sociological perspectives, highlighting factors like naive realism and confirmation bias, inherent in human nature. As shown in Fig. 1, social users, influenced by their own biases, tend to seek and accept information that confirms their beliefs, then share it. This forms collective opinions marked by cognitive bias, where the shared news tends to be consistently genuine or fake. From the perspective of fake news dissemination, social users who frequently repost the same news often reflect collective social opin-

ions, influenced by a specific cognitive bias. Furthermore, the veracity of news shared by these collectives typically remains consistent, being mostly genuine or predominantly fake [15]. We term this phenomenon as "social context veracity dissemination consistency".

Therefore, we propose DisCo-FEND, a social context veracity **Dis**semination **Co**nsistency-guided case-based reasoning augmentation for Fake News Detection (**FEND**). The primary research contribution of DisCo-FEND, is to organically integrate the "social context veracity dissemination consistency" with prompt-tuning to complete the FEND task in data-scarce scenarios.

For the incorporation of "social context veracity dissemination consistency" and our model, we constructed a Veracity Dissemination Consistency Network based on the collected social context-based case dissemination records. This network captures how news cases spread among users and identifies patterns in user interactions with multiple news items.

The Veracity Dissemination Consistency Network is utilized for DisCo-FEND inference, facilitating the retrieval of first-order and second-order relationships among neighboring news cases. The network comprises a multitude of news cases serving as nodes, with the relationships between news shared by the same social users forming connections or links. Each node represents a distinct news case, and the edges represent the dissemination pathways among users, highlighting both direct and indirect dissemination patterns. This allows for a detailed representation of how information flows and clusters within the social context.

During inference, we employed a novel "multiple FEND cases-based reasoning enhancement" strategy. This strategy optimizes the predictions of a news case by utilizing the prediction results from multiple news cases retrieved from Veracity Dissemination Consistency Network that have first or second-order social context dissemination relationships with the current news case. By assigning proportional weights, the prediction of current news case is further refined by integrating features from adjacent news cases.

To determine the practical effectiveness of this strategy, we conducted extensive experiments using real-world datasets. We evaluated DisCo-FEND on multiple benchmarks [8, 13] , comparing its performance against baseline models and state-of-the-art fake news detection methods [12, 10, 4, 15]. The results demonstrated a significant improvement in accuracy and consistency of predictions when employing the "multiple FEND cases-based reasoning enhancement" strategy.

The main contributions of this work can be summarized as follows:

1. We propose a veracity consistency case-based prompt learning approach, DisCo-FEND, which integrates social context veracity dissemination consistency into case-based reasoning. DisCo-FEND enhances FEND performance through a novel reasoning strategy based on the *Veracity Dissemination Consistency Network*.

2. We introduce a novel expanded label words-based verbalizer engineering in prompt-tuning, consisting of an improved expanded label words acquisition approach, and an enhanced multi-label words mapping strategy.

2 DisCo-FEND

In this section, we provide an overview of our Social Context Veracity Dissemination Consistency-guided Case Reasoning Augmentation for Fake News Detection, DisCo-FEND. DisCo-FEND addresses the problem of detecting fake news by examining full news articles as input, along with their associated social context. The task is framed as a binary classification problem, where each article is labeled as either true or false.

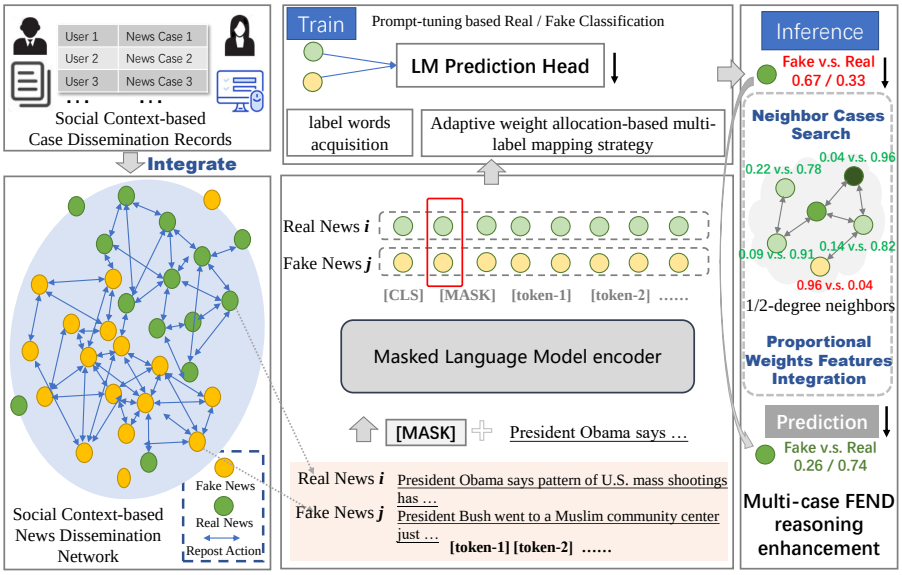


Fig. 2. The architecture of our DisCo-FEND, which is a veracity consistency case-based prompt learning approach. Utilizing the constructed social context veracity dissemination consistency network (left) to train the "cross-case embedding optimization" task; Leveraging the Masked LM-based pseudo prompt-tuning to conduct the primary task, Real/Fake binary classification for FEND (center); And the adopted multiple FEND cases-based reasoning enhancement strategy during the FEND inference phase (right).

2.1 Construction of Veracity Dissemination Consistency Network

In Section 1, we focused on analyzing the social phenomenon, confirmation bias, along with the fake news dissemination phenomenon: "social context veracity dissemination consistency".

Building upon "social context veracity dissemination consistency", we constructed a undirected graph, Veracity Dissemination Consistency Network. It is comprised of a multitude of real-world social news cases serving as nodes, with

the connections between news cases shared by the same users depicted as links among the nodes. It is utilized for the "multiple FEND cases-based reasoning enhancement" strategy during DisCo-FEND inference, streamlining the retrieval of first-/second- order neighboring news cases from the extensive pool of cases.

This network is built using the FEND dataset $\mathcal{D}$ and the social context-based case dissemination records $\mathcal{R}$. The Veracity Dissemination Consistency Network generates undirected graph $\mathbf{G} = (\mathcal{N}, \mathcal{L})$. The undirected graph consists of Nodes $\mathcal{N}$ representing news article cases and Links $\mathcal{L}$ denoting news cases reposting relationships. $\mathbf{G}$ illustrates the interconnections between these news cases, showcasing the sharing level through the frequency of reposting by the same social user.

For efficient retrieval of massive graph nodes and edges on the Veracity Dissemination Consistency Network, we utilize cutting-edge graph databases such as **Neo4j**[7] and **Nebula Graph**[8]. We opted for **Neo4j** in our experiment, an open-source tool known for its advanced storage technology, to ensure optimal read and write efficiency.

2.2 Prompt Learning-based Rumor Verification

The "prompt learning-based rumor verification" is the primary task that used for FEND (real/fake label binary classification).

Firstly, given a news article $\mathbf{S} = \{s_i\}_{i=1}^{N}$ composed of N words ($\mathbf{S} \in \mathcal{C}$), we manually construct a prompt template containing special [MASK] tokens and slot of news article $\mathbf{S}$, which is successive served as the prompt fed to PLM.

For the sake of simplicity, we utilize a single also simple fixed template $\mathbf{T}$: "It is a [MASK] news: [SLOT of $\mathbf{S}$]". By integrating the news article $\mathbf{S}$ with the prompt template $\mathbf{T}$, we obtain the PLM input $\hat{\mathbf{S}}$, which can be formulated as Eq. 1:

$$\hat{\mathbf{S}} = \mathbf{T}(\mathbf{S}) = It\ \ is\ \ a\ \ [MASK]\ \ news:\ \ \mathbf{S}. \tag{1}$$

$$\mathbb{V} = \mathcal{PLM}(\hat{\mathbf{S}}) = [v_{[CLS]}, v_{It}, ..., v_{[MASK]}, ..., v_{[SEP]}]. \tag{2}$$

As depicted in Eq. 2, taking $\hat{\mathbf{S}}$ as $\mathcal{PLM}$ input, BertForMaskedLM outputs a probability distribution of vocabulary words for each token in the textual input, represented by the sequence $\mathbb{V}$. Here, $v_i \in \mathbb{R}^{|\mathcal{V}|}$, where $\mathcal{V}$ denotes the length of the PLM vocabulary.

Given the vocabulary probability distribution $v_{[MASK]}$ at the token position of [MASK], the label word set denoted as $\mathbb{L}$, and $P(i, j)$ indicating the function to determine the probability of word j from logits vector i, along with the function $f(\cdot)$ which maps from $\mathbb{L}$ to $\mathbb{C}$, this reasoning process can be elucidated as follows:

$$\hat{\boldsymbol{c}} = f(\text{Argmax}_{\boldsymbol{l} \in \mathcal{L}}\, P(v_{[MASK]}, \boldsymbol{l}), \mathbb{L}). \tag{3}$$

[7] Neo4j: `https://neo4j.com/` *[Accessed on 2024.03].*
[8] Nebula Graph: `https://www.nebula-graph.com.cn/` *[Accessed on 2023.03].*

This process involves an "Argmax" search function to find the label word l with the highest probability score derived from the [MASK] token position's probability distribution $v_{[MASK]}$.

2.3 Expanded Label Words-based Verbalization Engineering

We further integrate *Expanded Label Words-based Verbalization Engineering* into our DisCo-FEND framework. The motivation for expanding label words comes from the inherent limitation of relying on a single label word for effective prediction, as this can lead to insufficient elicitation of relevant knowledge from PLMs. By broadening the set of label words, we can better capture diverse semantic nuances, improving the accuracy and robustness of the verbalizer in distinguishing between real and fake news.

Our proposed expanded label words-based verbalizer engineering consists of two strategies:

- 1. An improved approach for acquiring high-quality real/fake semantic-related label words;
- 2. An enhanced multi-label words mapping strategy based on dynamic attention mechanism and adaptive weight allocation.

Expanded Label Words Acquisition We employed various tools and techniques to search through external knowledge repositories, including WordNet, ChatGPT, Word Embedding, Masked LM, to filter high-quality label words [16, 6, 5]. Specifically, we used WordNet for synonym expansion, ChatGPT for contextual understanding, Word Embedding for semantic similarity, and Masked LM for contextual prediction. In our experiments, we adopted three approaches to obtain fake and real categories' label words:

WordNet SynSet Substitution: This technique involved manual retrieval through WordNet, a thesaurus that offers synonyms for any given word[9]; Specifically, we begin with the "real" and "fake" words as initial label words, and then we retrieve their corresponding synonyms sets, known as SynSets, from WordNet.

Word Embedding Substitution: We employed the Word Embedding to find words with semantic similarity as synonyms [1]; We employ both Word2Vec and GloVe techniques to seek words semantically (Euclidean distance or cosine similarity) akin to the initial label words.

MLM-based Masked Word Substitution: During MLM-based masked word substitution, we first applied coarse filtering to the label words using WordNet SynSet Substitution and Word Embedding Substitution. Then, we conducted fine filtering using MLM-based Masked Word Substitution to achieve finer selection.

[9] Princeton WordNet: `https://wordnet.princeton.edu/` *[Accessed on 2024.02].*

Enhanced Multi-Label Words Mapping Strategy Based on the expanded label words $\hat{\mathcal{W}} = D_{real}$ and D_{fake}, we devised an enhanced multi-label words mapping strategy in the verbalizer engineering of DisCo-FEND. We employ dynamic attention mechanism and adaptive weight allocation to provide a more stable learning process.

Specifically, for each category, such as the D_{fake} category, and for each word d_i^{fake} in D_{fake}, we assign a learnable weight w_i^{fake} to it. These weights form a vector for "fake" category, $\mathbf{w}^{fake} = [..., w_i^{fake}, ...]$, where $\mathbf{w}^{fake} \in \mathbb{R}^{|D_{fake}|}$ initialized as a zero vector. The weights $\mathbf{w}^j$ are normalized within each category j's label word set D_j as follows:

$$\alpha_{d_i^j} = \frac{\exp\left(w_i^j\right)}{\sum_{d_k^j \in D_j} \exp\left(w_k^j\right)}. \tag{4}$$

where $\alpha_{d_i^j}$ represents the normalized adaptive weight of the i-th label word d_i^j in category j; w_i^j and w_k^j refer to the adaptive weights of d_i^j and d_k^j, respectively, before normalization.

Utilizing the computed normalized adaptive weights for label words, we aggregate the weighted probabilities of all label words in each category. This aggregation involves calculating a weighted sum, and based on the resulting total weight for each category, we ascertain the ultimate category label for the FEND task. Throughout the training process, we anticipate that smaller weights will be learned for noisy label words, thereby minimizing their impact on predictions.

2.4 Multiple FEND cases-based reasoning enhancement Strategy

During the inference of DisCo-FEND, we introduce a technique based on cases-based reasoning, called the Multiple FEND cases-based reasoning Enhancement Strategy. This strategy retrieves neighboring news cases through the social context veracity dissemination consistency network, and integrates the veracity features of these cases to further optimize the detection results.

Given a news case to be predicted (denoted as $\mathbf{n}^0$), and the Veracity Dissemination Consistency Network $\mathbf{G} = (\mathcal{N}, \mathcal{L})$ stored in graph databases like Neo4j, we retrieve first-order nodes and links/relationships ($\mathbb{N}^1 = \{\mathbf{n}_0^1, ..., \mathbf{n}_i^1, ...\}; \mathbb{R}^1$, which are news cases), as well as second-order nodes and links/relationships ($\mathbb{N}^2 = \{\{\mathbf{n}_0^2, ..., \mathbf{n}_i^2, ...\}; \mathbb{R}^{21}; \mathbb{R}^{22}$, which are also news cases) adjacent to the current news case, using the following database query statement:

1. MATCH (n:NC{$\mathbf{n}^0$})-[rel1:repost] → (m:NC) RETURN rel1,m;
2. MATCH (n:NC{$\mathbf{n}^0$})-[rel1:repost] →[rel2:repost] → (m:NC) RETURN rel1,rel2,m;

where the "NC" denotes the type of News Case; the "repost" represents the link type; "rel1,m" in code # 1 denote the $\mathbb{N}^1, \mathbb{R}^1$; "rel1,rel2,m" in code # 2 denote the $\mathbb{R}^{21}, \mathbb{R}^{22}, \mathbb{N}^2$.

Since a node may simultaneously be a first-order and second-order node of the current news case $\mathbf{n}^0$, to prevent duplicate calculations, we deduplicate the

nodes in sets $\mathbb{N}^1$ and $\mathbb{N}^2$. Specifically, we retain all nodes in the first-order node set $\mathbb{N}^1$ and remove nodes from $\mathbb{N}^2$ that appear in $\mathbb{N}^1$.

Subsequently, we calculate the trustworthiness of each news case in the deduplicated node sets of $\mathbb{N}^1$ and $\mathbb{N}^2$. Trustworthiness reflects the confidence level in the veracity prediction results for each news case, ranging from 0 to 1, where higher values indicate greater reliability.

Given any news case from sets $\mathbb{N}^1$ and $\mathbb{N}^2$, denoted as m_i, and a trained prediction module from DisCo-FEND for Prompt Learning-based Rumor Verification, represented as the function $P(\cdot)$, where the input is the news case and the output is its real/fake label, i.e. (0 or 1); $P(\cdot)$ determines the news case veracity by simply selecting the category corresponding to the highest probability value. $Bool2Int(\cdot)$ is a function that converts boolean values to integers. The calculation process of the trustworthiness β_{m_i} of each news case is as follows.

Similarly, we first obtain the first-order and second-order neighboring news case sets of news case m_i using Code # 1, denoted as $\mathbb{M}^1$ and $\mathbb{M}^2$, respectively. Following that, we derive the trustworthiness of news case m_i through the following formula:

$$\beta_{m_i} = \frac{\sum_{m_j \in \mathbb{M}^1 \cup \mathbb{M}^2} Bool2Int(P(m_i) == P(m_j))}{|\mathbb{M}^1| + |\mathbb{M}^2| - 1}. \tag{5}$$

where the $|\mathbb{M}^1|$ and $|\mathbb{M}^2|$ respectively denote the length of $\mathbb{M}^1$ and $\mathbb{M}^2$.

After obtaining the trustworthiness β_{m_i} of each news case m_i in $\mathbb{N}^1$ and $\mathbb{N}^2$, we calculate the influence weight δ_{m_i} of each neighboring news case m_i in $\mathbb{N}^1$ and $\mathbb{N}^2$ on the current news case to be detected through the following steps:

$$\gamma_{m_i} = \begin{cases} \mathbf{R^1}(m_i) & if \quad m_i \in \mathbb{N}^1; \\ Min(\mathbf{R^{21}}(m_i), \mathbf{R^{22}}(m_i)) & if \quad m_i \in \mathbb{N}^2. \end{cases} \tag{6}$$

where the γ_{m_i} denotes the adopted news reposting frequency in the links/ relationships $\mathbb{R}^1, \mathbb{R}^{21}$, and $\mathbb{R}^{22}$ between the current news case $\mathbf{n}^0$ and m_i of $\mathbb{N}^1$ and $\mathbb{N}^2$; $\mathbf{R^1}(\cdot), \mathbf{R^{21}}(\cdot), \mathbf{R^{22}}(\cdot)$ denote the search function of news reposting frequency on each $\mathbb{R}^1, \mathbb{R}^{21}$, and $\mathbb{R}^{22}$, respectively.

Eq. 6 reflects the reposting frequency of neighboring news cases selection strategy. For second-order nodes, we consider taking the minimum reposting frequency from both the first- and second-order links, which is theoretically explainable from the decreasing entropy of news dissemination information.

$$\delta_{m_i} = \frac{log(\gamma_{m_i} \times \beta_{m_i})}{\sum_{m_k \in \mathbb{N}^1 \cup \mathbb{N}^2} log(\gamma_{m_k} \times \beta_{m_k})}. \tag{7}$$

where $log(\cdot)$ denotes the natural logarithm, used to transform the data for normalization purposes.

Eq. 7 illustrates that we consider both the adopted news reposting frequency γ_{m_i} and the trustworthiness β_{m_i} of news case m_i as proportional conditioning

factors for the influence weight δ_{m_i} on the current news case. Through the Softmax function, the values of these products are transformed into a probability distribution, thereby forming the magnitude of influence weight.

As depicted in Fig. 2, given the current news case $\mathbf{n}^0$, through the enhanced multi-label words mapping strategy in Section 2.3, we obtain the predicted probabilities for the real and fake categories respectively as $p_+(\mathbf{n}^0)$ and $p_-(\mathbf{n}^0)$.

Then, we set a threshold K, and for news cases where $|p_+(\mathbf{n}^0) - p_-(\mathbf{n}^0)| > K$, we default to selecting the category with the higher predicted probability as the final FEND prediction. For news cases where $|p_+(\mathbf{n}^0) - p_-(\mathbf{n}^0)| < K$, we incorporate a multiple FEND cases-based reasoning enhancement, which leverages news cases with higher dissemination consistency to further refine predictions.

The specific process for refining the predictions of the news case $\mathbf{n}^0$ is as follows:

$$\begin{array}{l}\left[\hat{p}_+\left(\mathbf{n}^0\right); \hat{p}_-\left(\mathbf{n}^0\right)\right] = \mu \times \left[p_+\left(\mathbf{n}^0\right); p_-\left(\mathbf{n}^0\right)\right] \\ +(1-\mu)\sum_{m_i \in \mathbb{M}^1 \cup \mathbb{M}^2}\left(\left[p_+\left(\mathbf{m}_i\right); p_-\left(\mathbf{m}_i\right)\right] \times \delta_{m_i}\right)\end{array}. \tag{8}$$

where the $[\hat{p}_+(\mathbf{n}^0); \hat{p}_-(\mathbf{n}^0)]$ are the refined predictions of news case $\mathbf{n}^0$. We additionally set a hyperparameter μ ranging from 0 to 1, reflecting the extent to which neighboring news cases influence the original predicted values.

Leveraging the neighboring news cases from the social context veracity dissemination consistency network based on Eq. 8, we further adjust the labels probability distribution of uncertain news case. We then select the category corresponding to the adjusted probability with the highest value as the final real/fake prediction label, thereby enhancing the performance of the FEND task.

3 Experiments

3.1 Experimental Setup

Dataset We evaluate our DisCo-FEND using three widely-used real-world benchmark datasets: FakeNewsNet [13], which comprises the PolitiFact and GossipCop datasets, along with the FANG dataset [8]. These datasets consist of news articles sourced from reputable fact-checking websites, accompanied by corresponding social user engagement data retrieved from Twitter, including repost user IDs. To maintain consistency in replicating results, we adopted the same data partitioning method and few-shot setting (with K-shot, where K varies from 16 to 128) as previous researchers [4, 10, 15]. We split the data into training and test sets for training purposes, maintaining a 1:1 ratio of fake news to real news.

Baselines We assess our DisCo-FEND against several prominent baselines, each representing "Train-from-Scratch" approaches (employing FEND task-specific neural architectures) and "PLM-based" approaches (aiming to mitigate label scarcity through rich pre-trained knowledge in PLMs). "Train-from-Scratch" methods include: 1. **dEFEND-variant** [12], a modified version of the hierarchical attention model dEFEND; 2. **SentGCN** and 3. **SentGAT** are graph-based

methodologies leveraging Graph Convolutional Network (GCN) and Graph Attention Network (GAT), respectively; 4. **SAFE-variant**, a variant of SAFE that excludes the visual component and incorporates a TextCNN-based module to process news article texts; 5. **GCNFN** employs deep geometric learning to model news dissemination patterns alongside textual node embedding features; 6. **GraphSAGE** [8] constructs a heterogeneous social graph comprising news articles, sources, and social users for fake news detection. "PLM-based" methods include 1. **BERT-FT** and 2. **RoBERTa-FT** [2] utilize BERT and RoBERTa alongside a task-specific MLP to predict the veracity of news articles; 3. **PET** [10] provides task descriptions to PLMs for supervised prompt-tuning via task-related cloze questions and verbalizers; 4. **KPT** [4] broadens the label word space with class-related tokens of varying granularities and perspectives. 5. Prompt-and-Align [15] is a prompt-tuning method for few-shot FEND that harnesses the pre-trained LMs and external social context topology knowledge.

To ensure consistency in experimental conditions, we choose BERT-base from Huggingface[10]. Specifically, the "BertForMaskedLM" model of "bert-base-uncased" serves as the backbone for prompt-tuning, whether in the baseline methods or our DisCo-FEND approach.

Hyperparameters We configured the few-shot settings to {16, 32, 64, 128}, with a total of three independent training epochs per round. The batch size used for each model update was set to 8, and the maximum token sequence length for the PLM input was set to 128. We set the default random seed to 0 and the learning rate to 0.00005. The hyperparameters threshold K and μ were set to default values of 0.35 and 0.5, respectively, which were determined as the optimal settings through experimentation. The chosen evaluation metric is accuracy.

3.2 Main Results

We tested our DisCo-FEND against other baselines for accuracy on the PolitiFact, GossipCop, and FANG datasets, as shown in Table 1. We classified the baselines into two groups according to the "Train-from-Scratch" and "PLM-based" categories of FEND methods.

From this table, it is clear that compared to these baselines, our DisCo-FEND surpasses all methods in accuracy, validating the effectiveness and superiority of DisCo-FEND that we propose. This demonstrates the necessity of introducing expanded label words-based verbalization engineering and the "multiple FEND cases-based reasoning enhancement" strategy into our model. This also indirectly reflects the significance of incorporating the “social context veracity dissemination consistency” phenomenon into the few-shot FEND task.

3.3 Ablated Studies

In this section, we discuss the effectiveness of each module of our DisCo-FEND and the rationale behind the selection of hyperparameters through a series of comparative experiments.

[10] Huggingface: `https://huggingface.co/google-bert/bert-base-uncased`.

Table 1. Performance comparisons (% Accuracy) of our DisCo-FEND and baselines on three benchmarks. Some baseline statistics are from their respective literature [15]. Bold numbers signify the best performance, underlined indicate second-best. Results marked with * are enhanced using "multiple FEND cases-based reasoning".

FEND	**PolitiFact**				**GossipCop**				**FANG**			
Approaches	16	32	64	128	16	32	64	128	16	32	64	128
dEFEND-variant	51.78	54.62	61.06	66.34	50.43	50.35	51.48	52.66	50.19	50.85	50.91	54.60
SentGCN	56.34	51.95	56.81	56.23	49.57	49.33	50.04	53.87	51.35	50.60	52.58	54.45
SetnGAT	56.09	52.99	55.64	58.36	49.54	50.11	50.52	54.65	51.04	51.59	54.09	56.34
SAFE-variant	57.71	54.52	61.91	68.36	51.35	51.20	53.39	57.40	51.90	53.01	55.53	58.35
GraphSAGE	57.87	60.45	68.72	77.56	51.73	54.10	63.93	69.65	53.32	56.22	58.53	60.49
GCNFN	55.37	61.86	70.06	84.46	52.18	53.85	62.29	71.89	52.43	56.08	60.12	61.85
BERT-FT	61.27	67.48	73.52	77.43	52.45	52.74	55.02	59.28	53.28	54.71	57.02	58.41
RoBERTa-FT	54.80	61.16	78.95	81.44	52.54	54.17	54.09	61.35	51.47	54.56	56.92	60.92
PET	64.24	68.09	79.44	80.49	53.69	55.11	59.78	62.97	55.76	56.46	59.13	59.79
KPT	68.27	70.21	80.40	83.16	53.95	54.70	60.12	62.19	56.87	55.79	60.36	61.38
Prompt-and-Align	85.30	83.73	86.79	89.45	61.15	70.59	74.62	81.60	59.42	60.48	62.37	64.37
DisCo-FEND	**88.70**	**88.92**	**89.37**	**91.84**	**65.28**	**73.74**	**80.91**	**83.84**	**61.37**	**64.12**	**68.70**	**74.31**

Superiority of our Enhanced Multi-label Words Mapping We discuss the effectiveness of our proposed enhanced multi-label words mapping strategy, which is based on a dynamic attention mechanism and adaptive weight allocation, hereinafter referred to as Adaptive Allocation (default). This strategy allocates reasonable weights to different label words under each category to improve model prediction. We introduce three simple mapping strategies from label word probability distribution to categories for comparison: 1. **Maximum**: Selects the category corresponding to the label word with the highest probability value across all categories as final prediction; 2. **Averaged**: Selects the category corresponding to the label word with the highest average probability value among all categories as final prediction; 3. **Median**: Selects the category corresponding to the label word with the highest median probability value among all categories as final prediction.

As shown in Table 2, our dynamic attention mechanism and adaptive allocation-based label words mapping strategy achieves the best performance across all three benchmarks at 16-shot setting. Following it, the Averaged words mapping strategy shows the next best performance, highlighting the effectiveness of the dynamic attention mechanism and adaptive weight allocation algorithm designed in our DisCo-FEND. We believe that the reason Averaged performs well is that its calculation incorporates features from all label words, making predictions more robust. In contrast, Maximum and Median have lower feature coverage

Table 2. Comparisons of DisCo-FEND variants that employ different label words mapping strategies.

DisCo-FEND Variants		PolitiFact (16)	GossipCop (16)	FANG (16)
Adaptive Allocation (default)	Prompting	**79.38**	**60.67**	**58.77**
	Case-based Reasoning	**88.70***	**65.28***	**61.37***
Maximum	Prompting	76.30	57.94	57.82
	Case-based Reasoning	86.36	62.49	59.90*
Averaged	Prompting	78.44	59.83	57.50
	Case-based Reasoning	88.56*	64.77*	59.15
Median	Prompting	72.68	54.28	55.53
	Case-based Reasoning	80.04	60.06	57.82

and lack representativeness, making them more susceptible to the influence of noisy data, resulting in poor performance.

Selection of Hyperparameters μ and K Here, we explore the influence of hyperparameters, i.e. the threshold K and the influence extent μ, on the performance of DisCo-FEND to determine the optimal parameter values.

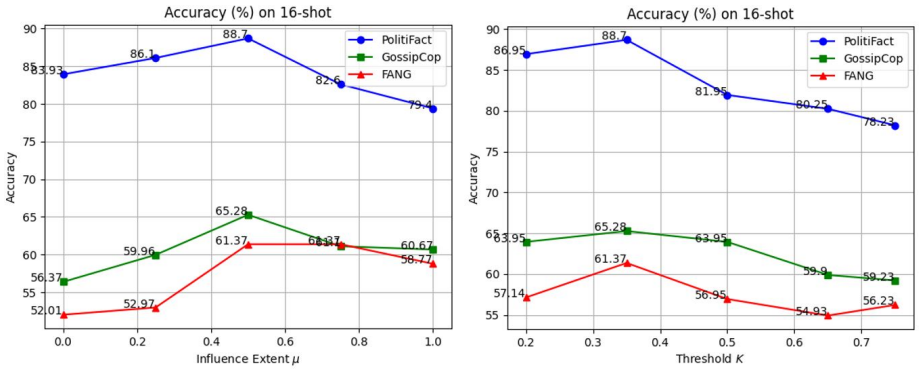


Fig. 3. The comparisons of DisCo-FEND that adopting different hyperparameter settings of μ and K.

Firstly, regarding the influence extent μ, we set six different values for μ, {0.0, 0.25, 0.5, 0.75, 1.0}. Here, $\mu = 0.0$ corresponds to DisCo-FEND relying entirely on the features of neighboring news cases from the social context veracity dissemination consistency network, while $\mu = 1.0$ corresponds to direct prediction through prompt learning without relying on multiple FEND cases-based reason-

ing enhancement. For the aforementioned ablation design, we conduct 16-shot experiments on three benchmarks. As shown in Fig. 3 (left), it can be observed that optimal performance is achieved when μ is between 0.4 and 0.6. This is also the reason why we chose $\mu = 0.5$ in our experiments.

Secondly, we explore the influence of threshold K, on the performance of DisCo-FEND to determine the optimal parameter values. We set five different values for K, $\{0.2, 0.35, 0.5, 0.65, 0.75\}$. This threshold K determines the extent to which the difference between the positive and negative probabilities of news cases affects the final FEND prediction. As shown in Fig. 3 (right), we conducted 16-shot experiments on three benchmarks for K. It can be observed that optimal performance is achieved when K is around 0.35. This is the reason why we chose $K = 0.35$ in our experiments.

3.4 PLM Bias Discussions for Domain-specific Fake News Detection

When using pre-trained models (PLMs) in domain-specific downstream tasks like fake news detection, bias is a critical concern. PLMs are typically trained on large, general datasets, which can lead to biases based on the distribution of topics or language present in the data. These biases can affect model performance when applied to specific domains, like fake news detection, where the context and nuances are unique. For instance, in fake news detection [4, 15], biases from pre-trained models can manifest in favoring certain language styles, sources, or topics, making them less effective in detecting fake news from domains or communities underrepresented in the pre-training data. Additionally, social phenomena like confirmation bias, where users share information aligning with their beliefs, exacerbate the challenge by creating echo chambers. Therefore, careful fine-tuning and techniques like case-based reasoning, as seen in DisCo-FEND, are necessary to mitigate bias and enhance model generalization across diverse datasets.

4 Conclusion

In few-shot FEND task, the weakness of supervised learning methods lies in the requirement of a large amount of labeled data. However, the construction of high-quality datasets is time-consuming and labor-intensive. In contrast, prompt learning technology has demonstrated superiority. Moreover, the theory of "user spreading news engagement bias" reveals the impact of social context on news dissemination, providing valuable insights. Inspired by this theory, we propose DisCo-FEND, a veracity consistency case-based prompt learning approach. It overcomes the limitations of previous supervised learning by incorporating the external knowledge of social context veracity dissemination consistency with the prompt-tuning technique. Our proposed DisCo-FEND framework represents a significant advancement in the research of few-shot fake news detection. Moreover, the introduction of the expanded label words-based verbalizer engineering further refines the understanding capability of our DisCo-FEND. The adoption

of the novel multiple FEND cases-based reasoning enhancement strategy further optimizes detection performance by leveraging the neighboring news cases from the social context veracity dissemination consistency network. Through extensive experimentation, our DisCo-FEND has been validated to effectively enhance the fake news detection capability in data-scarce scenarios. In conclusion, DisCo-FEND offers a powerful solution for combating misinformation, ensuring the integrity of information dissemination in the digital age.

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